To 3.2 or Not to 3.2? That is the Question

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EDDI 2013

“Whether ‘tis nobler in the mind to suffer the slings and arrows…”*

- There’s a new model driven DDI version already in the works. Shouldn’t I just wait for this?
  - The work on version 4 just started and while there will be packages coming out during the 18 month development period the full functionality of this won’t be available until the work is complete.
- There’s so many changes and I’ve got software for 3.1. Do I want to go through the pain of change?
  - Probably not. If what you have works, stick with it. But pay attention to some changes in 3.2 that will facilitate work with later models.
- What took so long? I really need 3.2 to do my work!
  - If you need the new functionality SWITCH!

**Shakespeare’s Hamlet (Act 3 Scene 1)*

There's so many changes and I've got software for 3.1. Do I want to go through the pain of change?

"And makes us rather bear those ills we have, Than fly to others that we know not of."*

- Bundling language equivalents in a single element
- Representations and missing values
- Managing identification content
- Name clashes
- New Code Value Types

**Shakespeare's Hamlet (Act 3 Scene 1)*

“And makes us rather bear those ills we have, Than fly to others that we know not of.”*

- Bundling language equivalents in a single element
  - Make sure multiple language content is complete
  - Note where cardinality has been constrained in 3.2
  - Loss of Identified Structured String – when multiple languages are involved
- Representations and missing values
- Managing identification content
- Name clashes
- New Code Value Types

**Shakespeare’s Hamlet (Act 3 Scene 1)*

"And makes us rather bear those ills we have, Than fly to others that we know not of."*

- Bundling language equivalents in a single element
- Representations and missing values
  - Valid and Invalid (Missing) Values are handled separately in 3.2 – and probably into the future
  - Create CodeSchemes that combine a CodeScheme of valid values and a CodeScheme of invalid (missing) values
  - Managed Representations (Numeric, Text, DateTime,…)
- Managing identification content
- Name clashes
- New Code Value Types

**Shakespeare's Hamlet (Act 3 Scene 1)*

“And makes us rather bear those ills we have, Than fly to others that we know not of.”*

- Bundling language equivalents in a single element
- Representations and missing values
- Managing identification content
  - Use URN’s! This is the only place where all the content of the full object identification can be expressed for Identifiable and Versionable types
- Name clashes
- New Code Value Types

**Shakespeare’s Hamlet (Act 3 Scene 1)*

“And makes us rather bear those ills we have, Than fly to others that we know not of.”*

- Bundling language equivalents in a single element
- Representations and missing values
- Managing identification content
- Name clashes
  - Note where object names are changing (these may be namespace specific)
- New Code Value Types

**Shakespeare’s Hamlet (Act 3 Scene 1)*

"And makes us rather bear those ills we have, Than fly to others that we know not of."*

- Bundling language equivalents in a single element
- Representations and missing values
- Managing identification content
- Name clashes
- New Code Value Types
  - Many attributes were changed to elements to support external controlled vocabularies (CodeValueType)
  - Constrain current content of these fields to values you plan to provide in an external controlled vocabulary

**Shakespeare's Hamlet (Act 3 Scene 1)*

What took so long? I really need 3.2 to do my work!

“'Tis a consummation Devoutly to be wished.”*

- Complex Questionnaires
  - Question Grids and Question Blocks
  - Data Processing
  - New representations and response domains
- Concept Management
  - Conceptual modeling
  - Managed representations and missing (invalid) values
- Content Management
  - Quality and authority statements
  - Reorganizing Organizations and Individuals
  - Archiving submitted content
- Consistency
  - Multilanguage content
  - Geographic content consistency
  - Scheme structure consistency

**Shakespeare's Hamlet (Act 3 Scene 1)*

Complex Questionnaires

- The primary reason for using 3.2
- Complex questionnaires are defined as those using:
  - Complex Question Grids
  - Question Blocks - where a set of questions are related to a specific stimulus item (common in educational testing)
  - Need for a broader set of response domains
    - Scales, capturing a mark on an object, geographic codes, ranking, distribution, etc.
  - Complex routing and reuse of data for calculations
    - Incorporating pre-loaded content
    - Extensive use of response information to inform the content of dynamic text
    - Data validation done during the collection process

3.1 Multiple Question Item

Did you visit any one of the following performances or facilities over the past 12 months?
(Circle the correct response for each)

Yes	No	a theatre performance
Yes	No	a cabaret performance
Yes	No	a concert of classical music
Yes	No	an opera or operetta
Yes	No	a concert of popular music, pop, jazz, musical or pop opera
Yes	No	a 'dance' event, houseparty
Yes	No	a ballet performance
Yes	No	the cinema
Yes	No	an art gallery
Yes	No	a museum

Simple Grid

15 At the time she left school, how many brothers and sisters did the patient have (living and dead)?

	Number still alive then	Number dead
Older than patient		
Younger than patient		

Source: British Perinatal Mortality Study (1958)

Example of a simple grid


```xml
<d:QuestionGrid versionDate="2013-11-20T12:09:32+00:00">
    <r:ID>qg1</r:ID>
    <d:QuestionGridName>
      <r:String xml:lang="en-GB">butler15</r:String>
    </d:QuestionGridName>
    <d:QuestionText>
    </d:QuestionText>
    <d:QuestionIntent xml:lang="en-GB">number of   siblings</d:QuestionIntent>
  <d:GridDimension rank="1" displayCode="false" displayLabel="false">
    <d:CodeDomain>
      <r:CodeListReference>
      <r:ID>cl1</r:ID>
      </r:CodeListReference>
    </d:CodeDomain>
  </d:GridDimension>
  <d:GridDimension rank="2" displayCode="false" displayLabel="false">
    <d:CodeDomain>
      <r:CodeListReference>
      <r:ID>cl2</r:ID>
      </r:CodeListReference>
    </d:CodeDomain>
  </d:GridDimension>
  <d:NumericDomainReference>
    <r:ID>n1</r:ID>
    <r:TypeOfObject>ManagedNumericRepresentation</r:TypeOfObject>
  </d:NumericDomainReference>
</d:QuestionGrid>
```


Source: British Perinatal Mortality Study (1958)

Complex Grid

SECTION III.—Please note carefully; the information in this section is to be got from records if at all possible. If this is not possible, get details from mother.

PAST OBSTETRIC HISTORY.—*Exclude present pregnancy.*

24 Has the patient had any previous pregnancies (including miscarriages)? Yes .... Y

No X

If " Yes " please give details below, taking the pregnancies in order of occurrence (the earliest first). Record twins as two separate births.

	Date of Delivery		Sex			Place of Delivery		Outcome of Delivery: Livebirth				Outcome of Delivery			Complications of Pregnancy					Method of Delivery				
Pregnancy Number	Month	Year	Male	Female	Birth Weight	Domiciliary	Institutional (including nursing homes)	Alive now	Died 28 days or later	Died 7-27 days inclusive	Died under 7 days	Stillborn	Miscarriage	Ectopic Pregnancy	Toxaemia	A.P.H.	Other complications	No complications at all	Not known whether any complications or not	Spontaneous	Forceps	Caesarean	Others	Method not known
1			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7
2			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7
3			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7
4			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7
5			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7
6			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7
7			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7
8			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7
9			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7
10			Y	X	........lbs. ....ozs.	0	1	2	3	4	5	6	7	8	Y	X	0	1	2	3	4	5	6	7

Source: British Perinatal Mortality Study (1958)

Question Grids with Roster

14. Please write down the names of the places you stayed overnight and the duration of the stays in chronological order separately.

Places you stayed overnight in chronological order			Type of accommodation for overnight stays (Please specify the number of nights)										
	Name of the place (City, town, etc.)	Total number of nights	Hotel	Motel	Holiday village	Pension	Camp/ Caravan	Youth And Holiday Camp	Own House	Rented House	House of a relative or friend	In the vehicle (Cruise, Yacht, Train, TIR)	Other (Please specify) (..............................)
			01	02	03	04	05	06	07	08	09	10	11
……	…………..	……………	………	………	………	………	……	………..	………	………	………	………..	……………………………….
……	…………..	……………	………	………	………	………	……	………..	………	………	………	………..	……………………………….
……	…………..	……………	………	………	………	………	……	………..	………	………	………	………..	……………………………….
……	…………..	……………	………	………	………	………	……	………..	………	………	………	………..	……………………………….

(Go to Question 15)

(Go to Question 17)

Source: TURKSTAT Tourism Survey

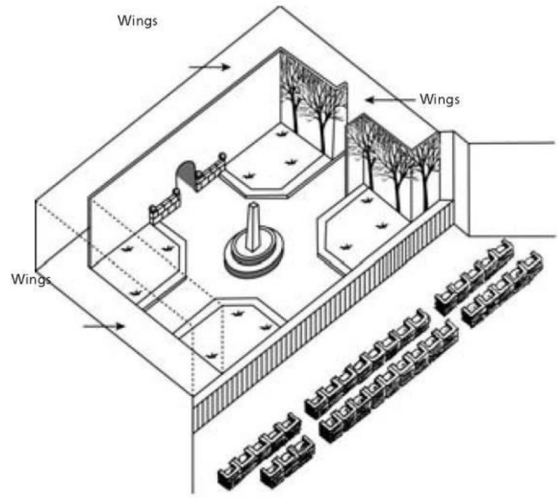
Question Block

2

QUESTION 9.4

The director positions the actors on the stage. On a diagram, the director represents Amanda with the letter A and the Duchess with the letter D.

Put an A and a D on the following diagram of the set to show approximately where Amanda and the Duchess are when the Prince arrives.



QUESTION 9.5

Towards the end of the extract from the play, Amanda says, "He didn't recognise me…".

What does she mean by that?

A. That the Prince didn't look at Amanda.

B. That the Prince didn't realise that Amanda was a shop assistant.

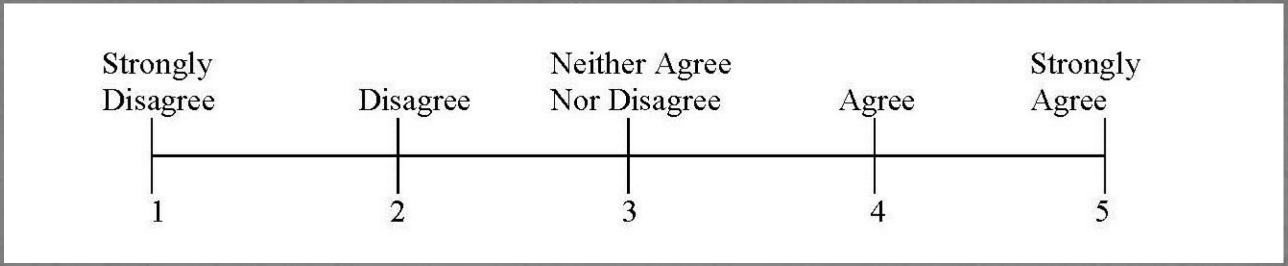
C. That the Prince didn't realise that he'd already met Amanda.

D. That the Prince didn't notice that Amanda looked like Léocadia.

TAKE THE TEST: SAMPLE QUESTIONS FROM OECD'S PISA ASSESSMENTS - ISBN 978-92-64-05080-8 - © OECD 2009

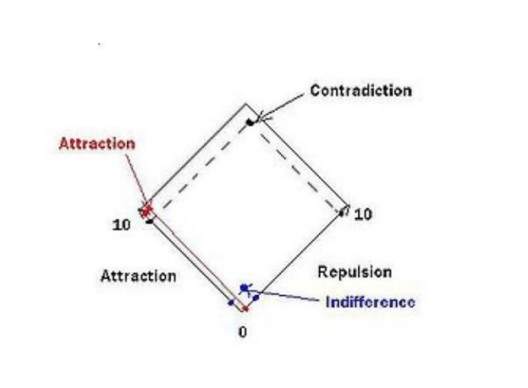
More Response Domains

- Location of marks on an image or recording
- Nominal level (for grids – checked or unchecked)
- Ranking (place categories in a specific order)
- Distribution (distribute a total value among a number of categories)
- Direct use of Geographic Structure and Location Codes
- Scales (anchored, unanchored, with/without a continuous line, etc.)



I am aware of the presence of God or the Divine

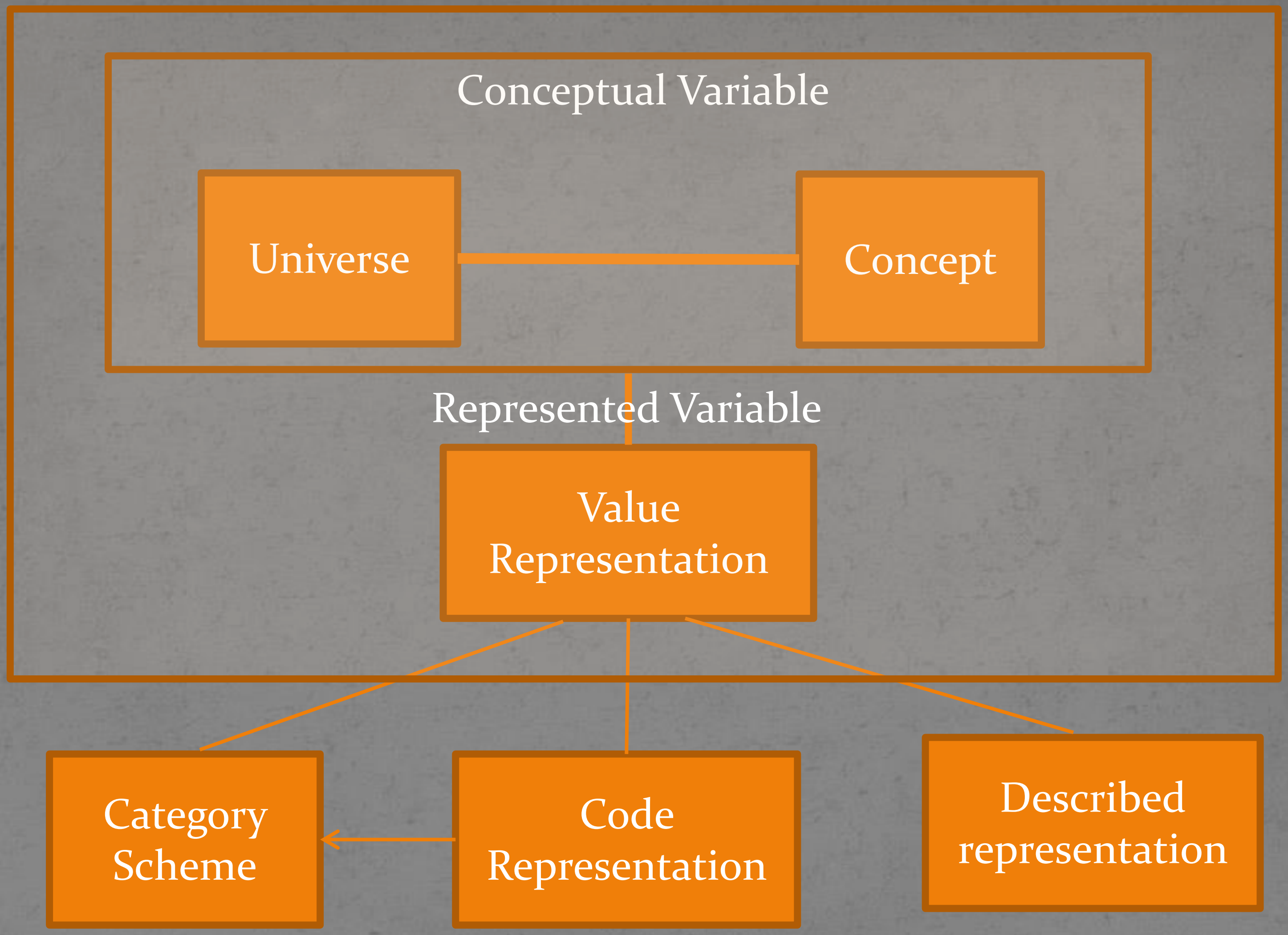
Never										Continually
0	1	2	3	4	5	6	7	8	9	10



Scales

Concept Management

- DDI is not fully there but 3.2 dramatically improves support for concept management and expression
- Alignment with GSIM content regarding Concepts, Conceptual Variables, and Represented Variables
- Managed representations allow numeric, textual, datatime and other representations to be captured once and reused
  - BONUS: managed representations can be mapped for comparison, i.e. a numeric representation of age to a coded set of age cohorts


Conceptual Variable
Universe
Concept
Represented Variable
Value Representation
Category Scheme
Code Representation
Described representation

Content Management: Quality

- Capture information on Quality Frameworks, Quality Standards, and other means of assuring the quality of the data, metadata, and process.
- Quality statements can be referenced from many levels within DDI
- Additional content to reflect authorization, legal status, budget, and evaluation procedures have been added

Content Management: Actors

- Organization Scheme content has been cleaned up and stripped down
- Organization and Individual information can be better managed over time (specific contact information is time stamped, etc.)
- Relationships between any two Organizations and/or Individuals is tracked separately under "Relation". Roles are assigned as part of the Relation
- Locations in DDI where names are associated with specific roles within the context of the Study can now reference an Individual or Organization

Content Management: Archiving

- Supports inclusion of all major packaging types as submitted (deposited) content (StudyUnit, Group, ResourcePackage)
- Supports Local Added Content of all packaging types
- Links Local Added Content to Submitted (Deposited) content by mapping
  - Local Object Reference
  - Depository Object Reference
  - Relationship Action (CodeValueType) i.e. Overides, Deletes, Extends, Restricts, etc.
  - Description

Language Consistency

- Bundling equivalent content expressed in multiple languages

```
<r:Label xml:lang="en">What is it?</r:Label>
<r:Label xml:lang="de">Was ist das?</r:Label>


<r:Label>
   <r:String xml:lang="en">What is it?</r:String>
   <r:String xml:lang="de">Was ist das?</r:String>
</r:Label>
```

Geographic content consistency

- Added composite geographies (Metropolitan Areas – composite of counties)
- Improved management of geographic change over time
- Added ability to use geographic structures (names and codes) and geographic locations (names and codes) directly as representations or response domains
- Added ability to reference specific locations for the purpose of defining Spatial Coverage

Scheme structure consistency

- All schemes have the following structural features
  - XxxxSchemeName, Label, Description
  - Inclusion of a scheme by reference
  - Object type included in-line or by reference
  - Object Group included in-line or by reference
- All schemes have an XxxxGroup to support grouping for administrative or conceptual purposes
  - XxxxGroupName, Label, Description, and TypeOfXxxxGroup
  - Universe and Concept references
  - Subject and Keyword
  - Group and object included by reference

What if I'm just starting out?

- A number of new tools are already available for 3.2
  - Colectica
  - Rogatus Platform
- If DDI-C or DDI-L 3.1 does it for you and there are tools to support your work, USE THEM
  - But….pay attention to the consistency changes made in 3.2 to make your work easy to migrate forward. These changes were made in anticipation of future version needs.

Questions?

Technical Committee – ddi-srg@icpsr.umich.edu